

AMBIENT SYSTEMS

Climate At Your Service

www.ambientsystems.ai

Problem

Real estate owners and operators struggle to optimize their facilities.

1. Limited Resources

It’s hard to find highly trained and experienced building operators who can optimize multiple facilities.

2. Technological Complexity

Many buildings operate using outdated legacy systems and equipment that lack interoperability.

3. Competing Priorities

Different buildings have different system requirements based on the unique needs of occupants and operators.

Specialized buildings require
more than energy management

Indoor Farms

Parameter Schedule
Harvest Prediction

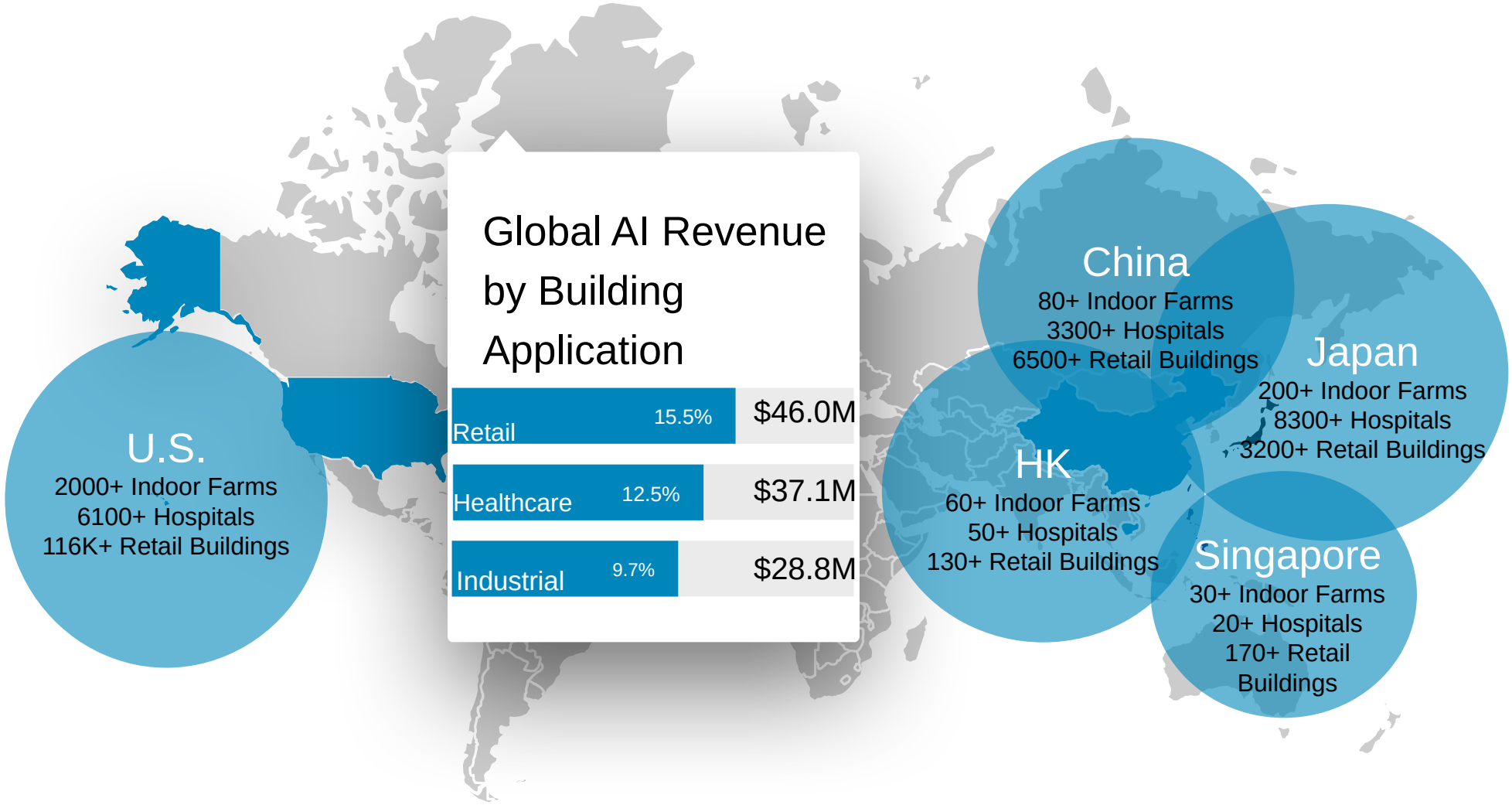
Hospitals

Zone Control Bacteria
Detection

Retail Stores

Customer Experience

Unlocking USD$110M of revenue across specialised segments



AI Platform Backed by IP





Sense

MVP sensors co-designed with manufacturing partners



Optimize

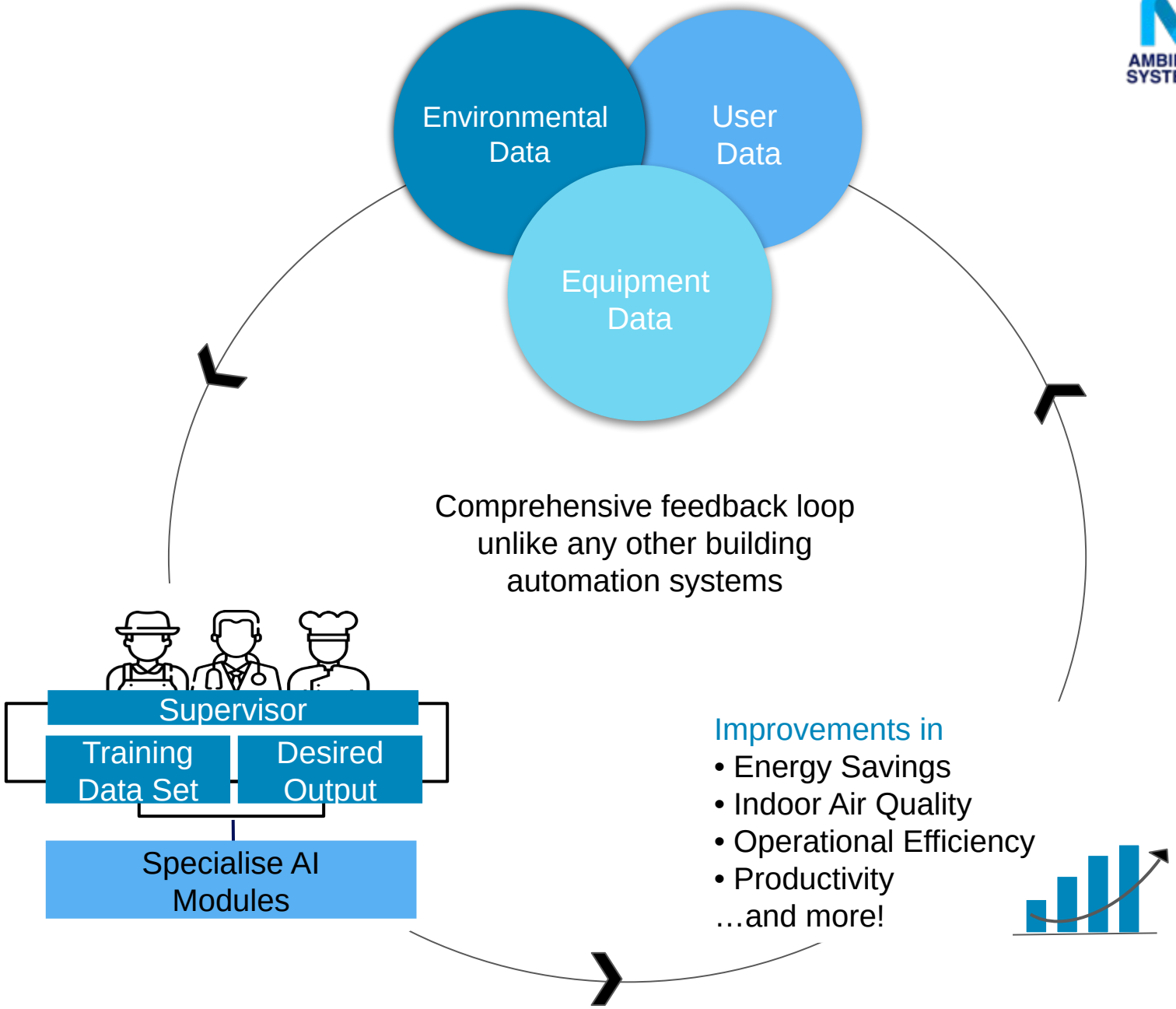
NUS National University of Singapore

Patent Pending Invention from National University of Singapore (NUS)

Control

Offering available after full-stack development

Building Optimization beyond Energy Savings



One step ahead with AI-enabled prescriptive maintenance

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Artificial Intelligence	Machine Learning	Deep Learning
		Datapoints gathered 
Automated Offerings • Building Performance Optimisation • Building Process Optimization	**Smart Offerings** • Asset Optimization • AI-based • AI-based Lighting Control	**Cognitive Offerings** • Predictive Maintenance

Identify general issues

Understand energy consumption by analysing primary datapoints

Personalised control

Understand energy flow and building occupancy

Collect external datapoints

Indoor Farming Overview

Improve crop yield in a market where only
51% of Indoor Farms are profitable

24% reduction in loss of crops due to
poor growing environment

Real-time improvements in crops efficiency
resulting in **0.5 - 1.5 years** payback period



Indoor farm operators
seeking to optimize crop performance
through customized insights across key
parameters.

Healthcare Overview

reviewing jerry's research

Hospitals looking to improve patient satisfaction and automation of equipment management.

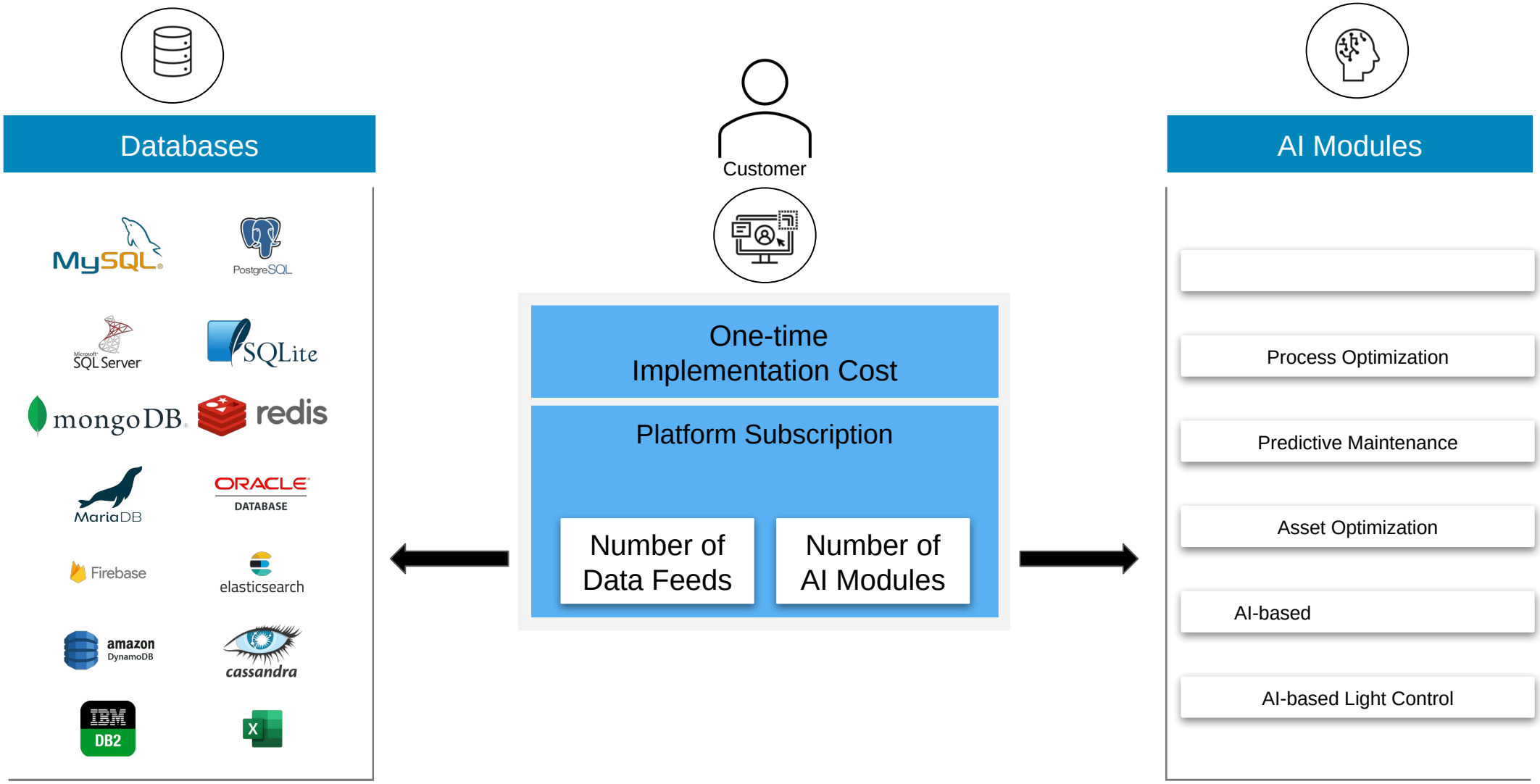
Retail Overview

reviewing jerry's research

- Reducing energy wastage and improving efficiency
- Intelligent inventory management and forecasting
- Optimising customer shopping experience
- Improving management efficiency and reducing labor cost

Retailers looking to optimize customer experiences,
forecasting, inventory management.

B2B SaaS Model



Usage-based Pricing Plan

pending from ivan

One-time Implementation Cost

Installation of sensors and equipment integration

by Project Basis

Basic

SGD$XX

- Insert Description

XX Data Feeds/mo
XX AI modules/mo

Advanced

SGD$XX

- Insert Description
- Insert Description

XX Data Feeds/mo
XX AI modules/mo

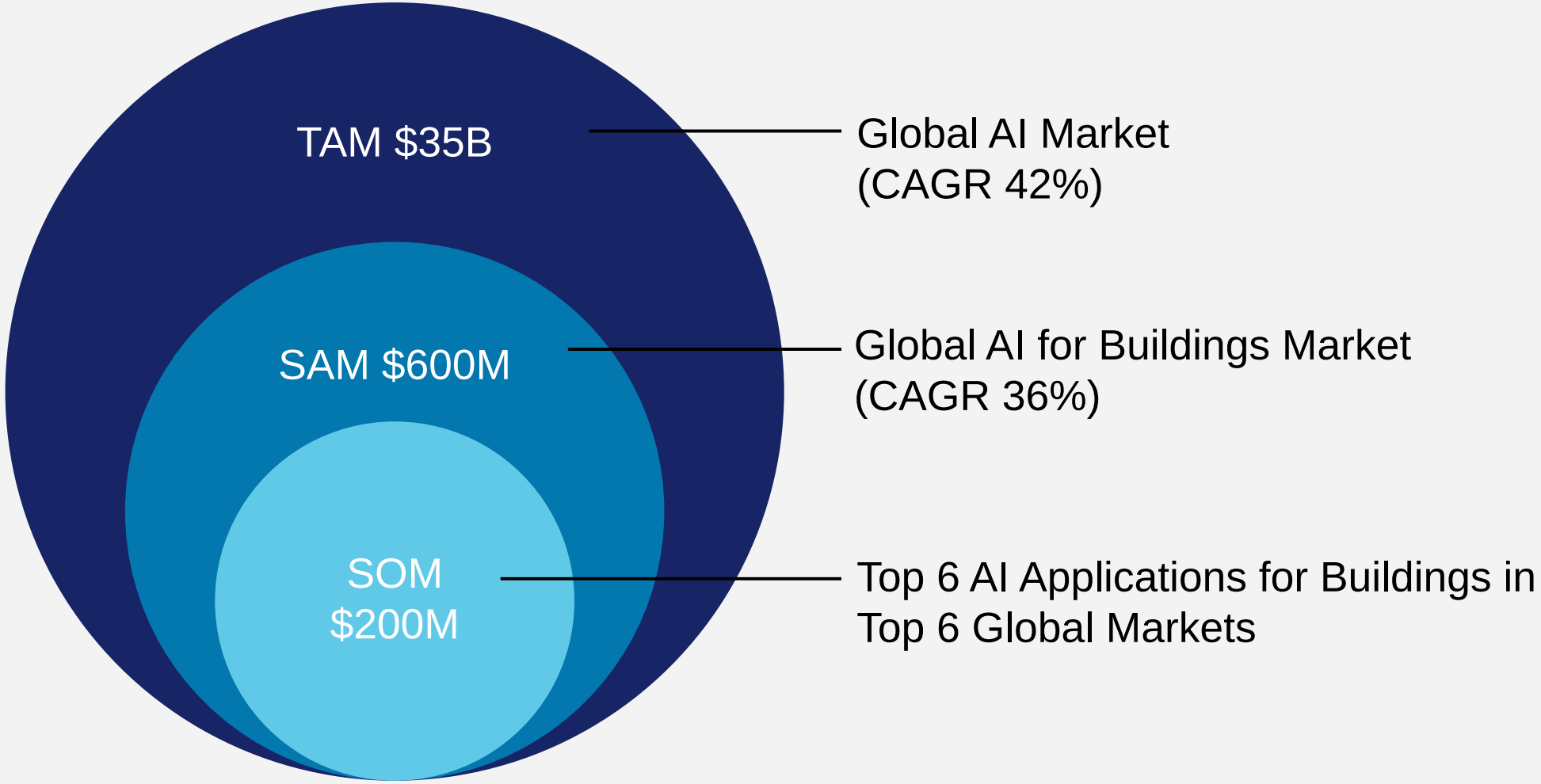
Premium

SGD$XX

- Insert Description
- Insert Description
- Insert Description

XX Data Feeds/mo
XX AI modules/mo

Opportunity



Source: Frost & Sullivan

Go-to-market Strategy

Monitor

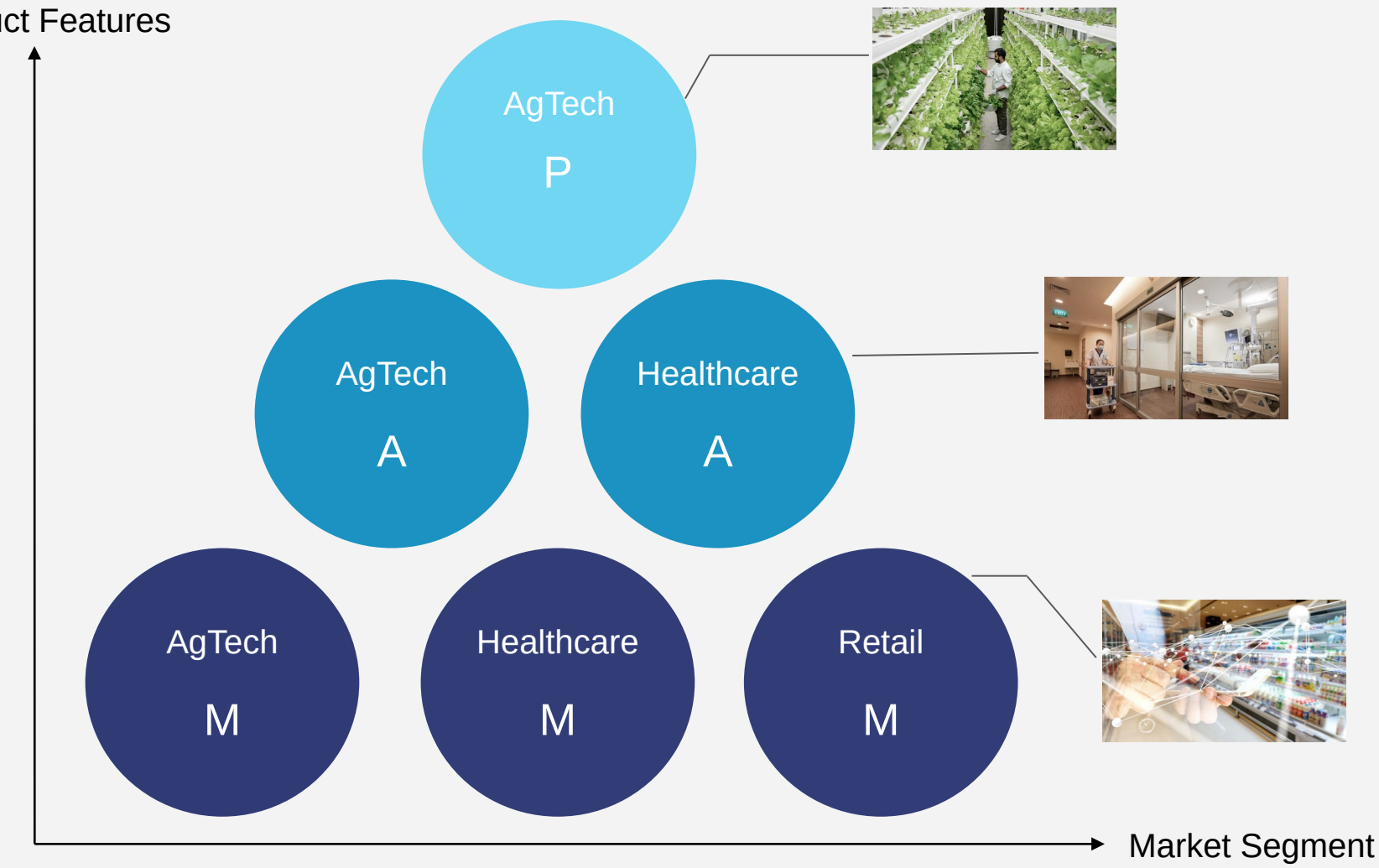
- Real-time data
- Online dashboards
- Push Alerts

Analyze

- Full HVAC connectivity
- Real-time environmental adjustments

Prescribe

- Integrate user data
- Obtain actionable insights according to needs



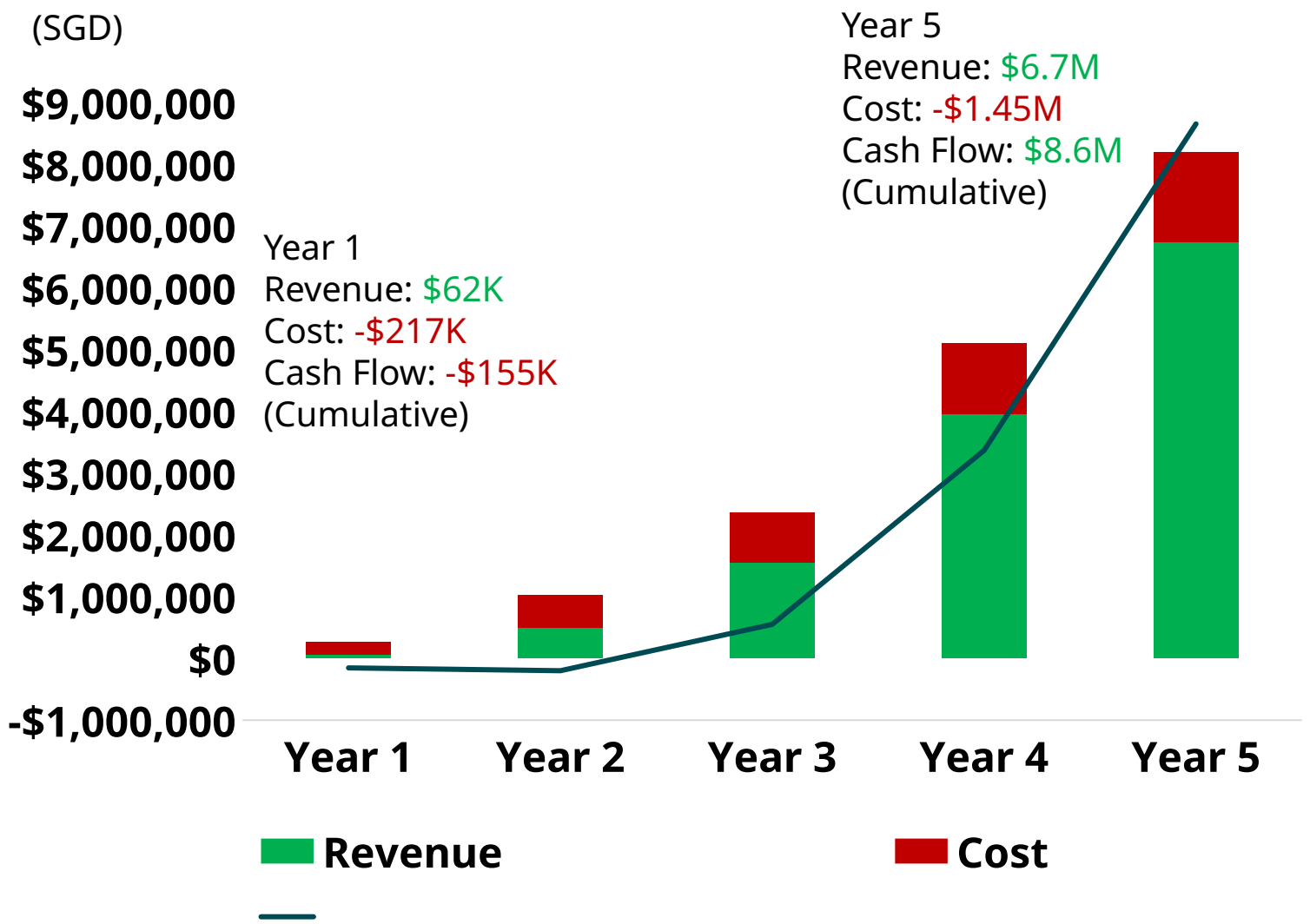
Traction & Current Milestones

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	2021									2022								
	Q2 2021			Q3 2021			Q4 2021			Q1 2022			Q2 2022					
	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep
Product	MVP v01				MVP v02					MVP v03								
Customers	LOI #1		LOI #2			Healthcare Offering					Retail Offering							
Fundraise			NUS GRIP			Pre-Seed Fundraising					Seed Stage Fundraising							
Markets																		

Revenue Forecast

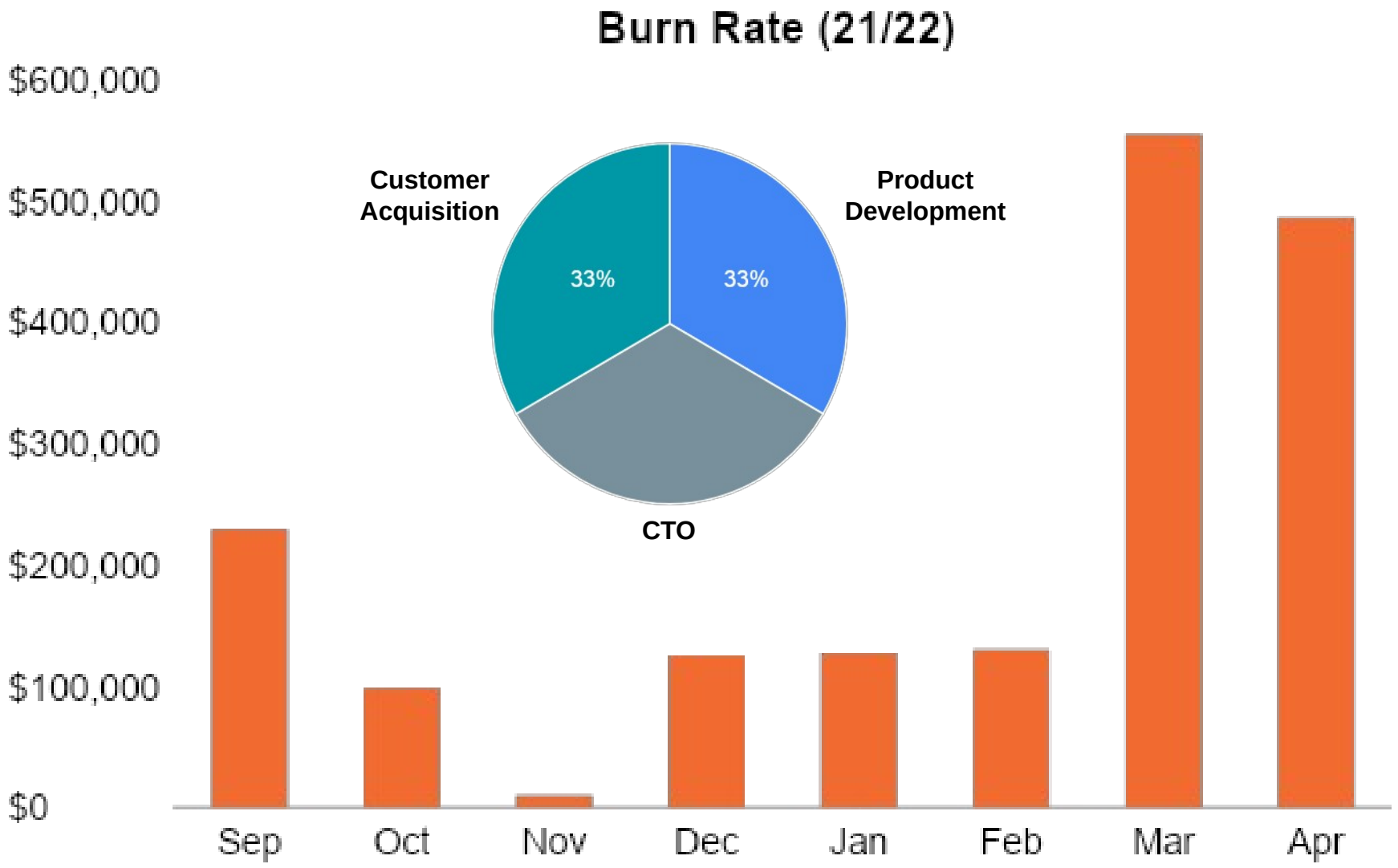
pending from ivan



81.2%
Average Gross Profit

$8.5M
NPV

Investment Ask and Use of Funds



pending from Ivan

SGDXXK

over X Months

Our Team

CEO

Background in Engineering and Computer Science

CSO

Experience in fundraising and strategy at APAC Unicorn startup

Design Lead

Creative Technologist with background in visual communications

Project Management Intern

Marketing at Unilever China

Developer

Built websites and applications for top brands in APAC

Thank You

Visit us at: www.ambientsystems.ai

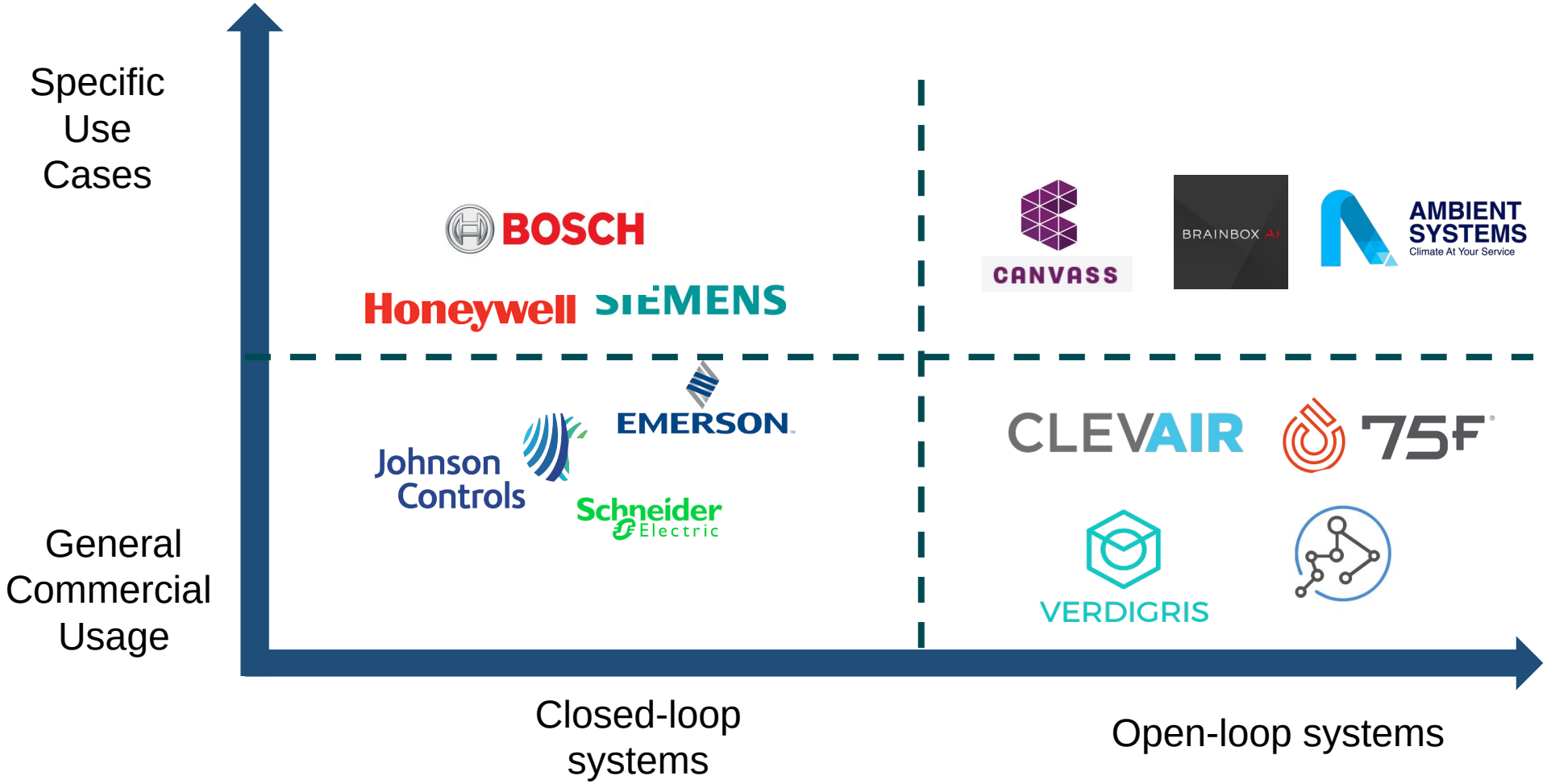
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Appendix

Competitive Analysis

ent provides value beyond energy savings



Source: Honeywell

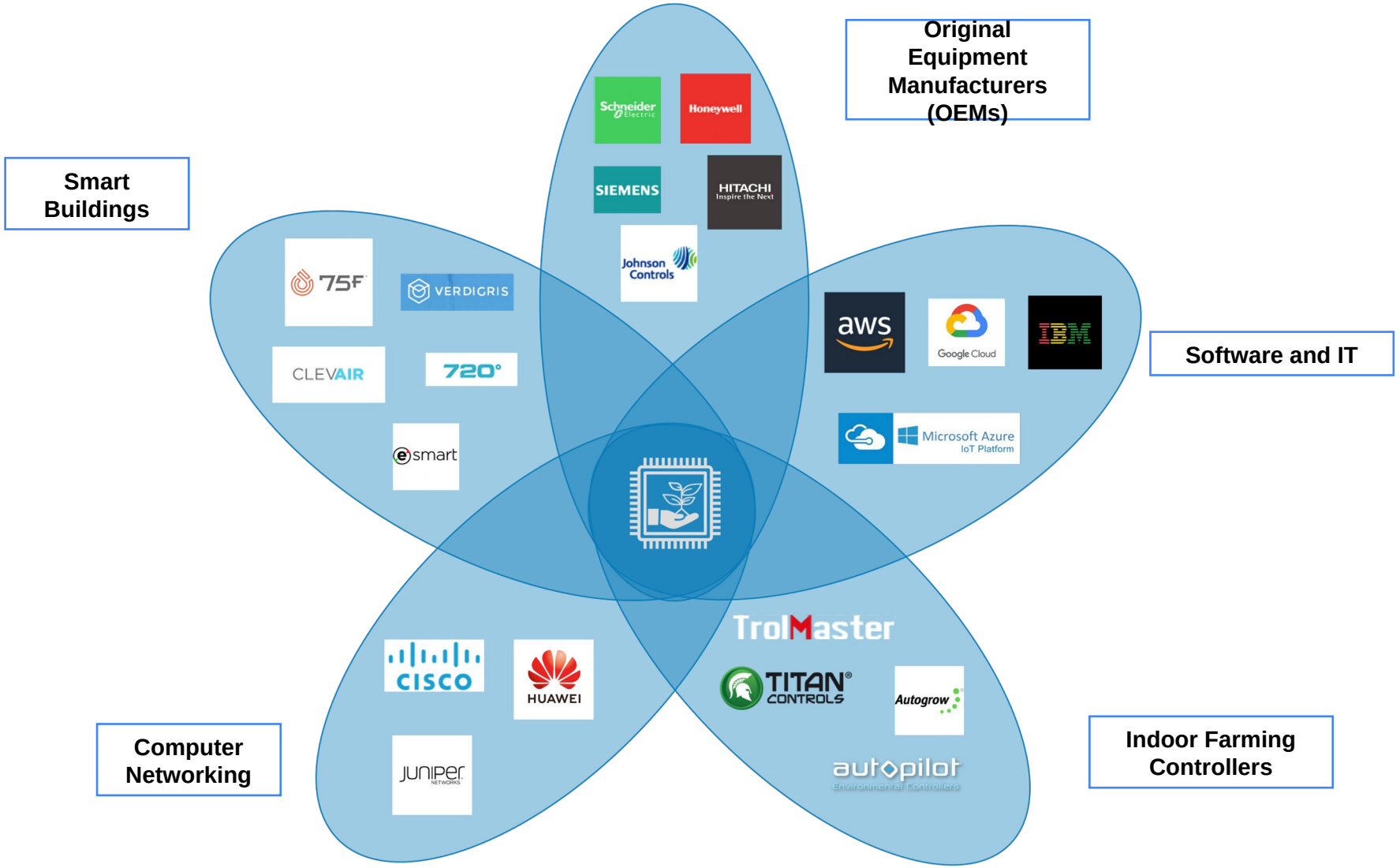
Competitive Analysis

nt provides end-to-end climate control solutions filling the gap between Building Automation and Indoor Farming systems

	AMBIENT SYSTEMS Climate At Your Service	Autogrow	75F	Johnson Controls	C3.ai	EPOS
Segment	**Multiple**	**Indoor Farming Controller**	**Smart Buildings**	**OEM**	**Hospitals**	**Retails**
Indoor Farming Features	**Yes**	Yes	No	No	No	No
Modern Web-based UI	**Yes**	No	Yes	Some	Some	Some
Wireless IoT Network	**Yes**	No	Yes	Yes	Yes	Yes
Advanced Optimization Methods	**Machine Learning**	No	Some	Some	Some	Some

Competitive Analysis

petitive Landscape: Turnkey Solution Provider



Competitive Analysis

Top Alternatives: Asia and Global

Company	Product	Link
Siemens	MindSphere	https://siemens.mindsphere.io/en
Honeywell	Forge	https://www.honeywell.com/us/en/honeywell-forge/buildings
	Metasys	https://www.johnsoncontrols.com/building-automation-and-controls/building-management/building-automation-systems-bas
Schneider	EcoStruxure	https://www.se.com/sg/en/work/campaign/innovation/overview.jsp
Bosch	Connected Buildings	https://blog.bosch-si.com/projects/bosch-singapore-campus-smart-building-concept-turned-reality/
EcoEnergy Insights	Cortix AI	https://www.cortix.ai/ https://ecoenergyinsights.com/how-we-do-it/proactive-solutions.php#how_we_do_it_video
75F	Facilisight	https://www.75f.io/solutions/products/multi-site-building-automation-75f-facilisight
BrainboxAI	BrainboxAI Dashboard	https://www.brainboxai.com/technology/
BuildingIQ	5i Intelligent Energy Platform	https://buildingiq.com/products/5i-intelligent-energy-platform/
Acuity Brands	Atrius	https://www.acuitybrands.com/brands/internet-of-things/atrius https://www.distech-controls.com/ https://www.acuitybrands.com/products/building-management/unified-building-control
Canvas AI	Canvas AI	https://www.canvass.io/platform

Competitive Analysis

Mapping of Companies Offering AI Applications in the H&B Industry

Key participants offering AI applications.

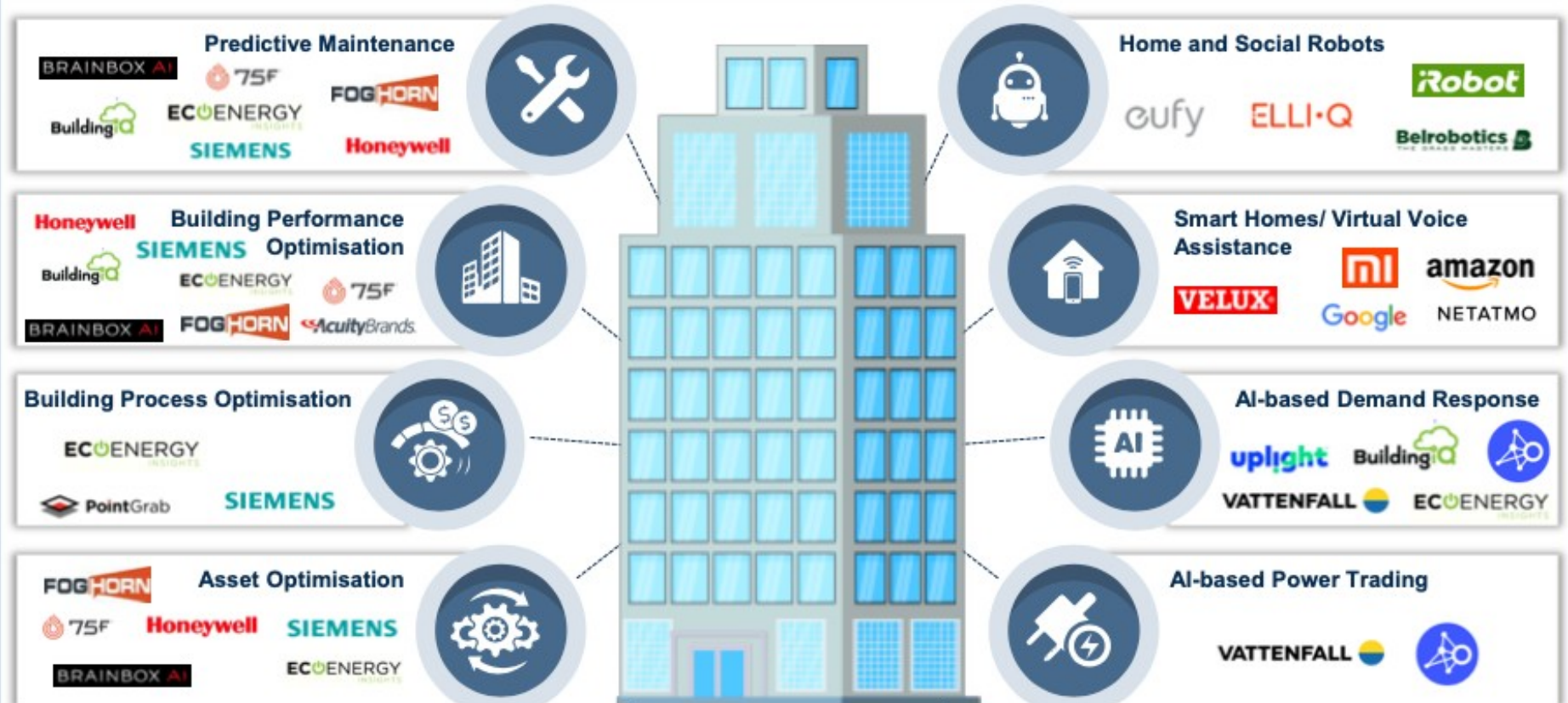


Image source: Freepik.com

Source: Frost & Sullivan

Competitive Analysis

Top AI Companies to Watch

Leading participants of building technologies.

Company	Growth	Innovation
SIEMENS	Siemens Smart Infrastructure drives the digital transformation of the buildings industry by creating open-standards-based Building Information Modelling (BIM)-ready products and services and by connecting its products and installed base to the new digital offerings of Siemens Comfy and Enlighted in the smart space area. Siemens' global team of data scientists are involved in the development of a next-generation AI platform for data-driven applications as a part of the company's digital services.	Siemens' MindSphere platform collects data from numerous factory and industry equipment and provides digital solutions such as remote diagnostic services, performance optimiser, remote maintenance, and digital whiteboard. The company might leverage this experience and develop a new data-driven application for smart buildings in collaboration with IBM.
Honeywell	Honeywell's building technologies (HBT) business in the non-residential segment is expected to continue its strong performance in the commercial fire safety domain and record modest growth in security solutions. Honeywell, after the spin-off of its homes business, Resideo, is well-positioned to grow profitably, with a 23% margin target by 2023.	Honeywell has announced the launch of Honeywell Forge Energy Optimization, a cloud-based, closed-loop, machine learning solution that studies the pattern of building energy consumption and continuously optimises building assets for improved energy efficiency without compromising on occupant's comfort.
Signify	Signify's light-emitting diode (LED)-based sales reached 78% of its total revenue, growing at a rate of 2.7% in 2019. LED-based sales are estimated to account for 85% of Signify's total sales in 2020, and its connected light points installed base is expected to exceed 65 million in 2020.	Signify will continue to strengthen its growth in data-enabled services by delivering IoT-based lighting solutions. Indoor and outdoor segment customers will benefit from the Interact IoT platform that leverages AI to deliver cost and operational efficiencies.
AcuityBrands.	Acuity Brands is the leading participant in the LED and non-LED luminaire market in the Americas, with 85% of its sales coming from the non-residential segment and 15% from the residential segment.	Acuity Brands' integrated package includes the Atrius sensory network, the building OS platform, and the Atrius platform that has been deployed for 320 million sq,ft. of space.

Source: Frost & Sullivan

Competitive Analysis

Top AI Companies to Watch (continued)

High-performing and emerging companies.

Company	Growth	Innovation
ECOENERGY INSIGHTS	For EcoEnergy Insights, 2019 was an exceptional year in terms of both organic growth and product innovation. Frost & Sullivan recognised the company's efforts and awarded it for demonstrating growth, innovation, and leadership in the global BEMS market. The company is expected to continue its strong performance in 2020.	EcoEnergy Insights' CORTIX platform has been upgraded to serve 19 categories of building equipment in different end-user verticals. Overall, 150,000 equipment are connected to the CORTIX platform. The platform's base engine is AI- and ML-driven, providing Insights-to-Action and, thereby, creating value for customers. More updates are expected in 2020.
FOGHORN	FogHorn ensures that its current product line and future road map remain relevant to its customers. The company focusses on improving the performance value for clients through an efficiency value play strategy. FogHorn has a dedicated global customer service team, with strong presence in Japan, Korea, Saudi Arabia, and the United States.	FogHorn leverages its industry-leading edge analytics capabilities by introducing features to its smart building AI solutions that include air quality monitoring, facial recognition with deep learning, video-based security surveillance for increased comfort and safety, and early water pump failure detection through sensory functionality.
75F	75F raised $18 million in Series A funding in 2019, led by Breakthrough Energy Ventures and Oil & Gas Climate Initiative. This funding will help the company invest in research and development (R&D) for product innovation and strengthening its partner network in North America, Middle East, and Asia-Pacific (APAC). The company expects to register more than 100% growth in 2020.	75F's self-learning building intelligence system improves a commercial building's operational efficiency and occupant experience by proactively controlling the HVAC, indoor air quality, and lighting settings based on inputs from sensors, weather data, and configuration settings and predictive algorithms powered by machine-to-machine (M2M) engine and cloud-computing data storage and processing power.

Source: Frost & Sullivan

Competitive Analysis

Top AI Companies to Watch (continued)

High-performing and emerging companies.

Company	Growth	Innovation
BRAINBOX AI	BrainBox AI has been growing at lightening speed. The company, which was working with less than half a million square feet in early 2019, hit 10 million by the end of the year. Currently, it manages around 18 million sq.ft. of space including retail spaces, offices, hotels, schools, warehouses, datacentres, hospitals, and grocery stores.	BrainBox AI's Artificial Intelligence engine uses advanced predictive control strategies with ML techniques that can self-learn a building's operating behaviour and energy flows and analyse and instruct the HVAC system to operate more efficiently, thereby reducing operating costs and carbon footprints significantly.
uplight	Tendril and Simple Energy merged as Uplight, with 75 North American utilities as customers. Earlier, Tendril had been on an acquisition spree, taking over EEme, an energy disaggregation start-up; EnergySavvy, a residential customer engagement software company; and FirstFuel Software, an energy efficiency data analytics company.	Uplight's combined business capabilities is a three-step process: engage, activate, and orchestrate. The first step is covered by Tendril's customer home energy reports and online interactions; the second step is covered by Simple Energy's software tools and management platforms to convert passive customers to active customers; and the third step is covered by Tendril's data analytics capability.
sense	Sense raised $18 million in fresh funding from investors led by Schneider Electric and including Shell Ventures, iRobot, Energy Impact Partners, Prelude Ventures, and Capricorn Investment Group. The company has partnered with 50 solar installers in more than 24 states in the US and Canadian provinces where home owners are embracing solar power.	Sense uses high-resolution data to detect small changes in magnitude, phase, and frequency and then incorporates machine learning algorithms to distinguish one appliance from another. The Sense Home Energy Monitor will be compatible with Google Assistant and the Belkin Wemo Insight Plug and TP-Link Kasa HS110 smart plugs in 2019.

Source: Frost & Sullivan

Competitive Analysis

Top AI Companies to Watch (continued)

High-performing and emerging companies.

Company	Growth	Innovation
BuildingIQ	BuildingIQ has been recording a CAGR of more than 20% since 2015 in the global BEMS market. Of late, the company has been busy expanding its partner network. BuildingIQ has raised a total of $6.5 million since it announced its capital raising in 2017. The company has an established market presence in the US, especially for large building portfolios.	BuildingIQ provides energy management solutions through its 5i energy intelligence AI platform for a wide range of buildings. Its solutions include a suite of services such as visualisation, building operations, and predictive control. The platform learns and evolves with existing building management systems and customer needs.
PointGrab	The Israeli smart building management company, PointGrab is an emerging firm that has been growing at an accelerated rate over the past few years. The company is backed by Signify, ABB Technology Ventures, and Mitsubishi Capital. It partners with Gooee for smart lighting, Serraview for connected workplace, and Tyco (now part of JCI). PointGrab has access to a large customer base through its partners.	The CogniPoint AI-based smart sensing platform is a combination of occupancy sensors and cloud management system. The occupancy sensors continuously send workplace occupancy data to the cloud management system, which in turn shares the data with a third-party building management system for overall building performance optimisation.
smappee	The Belgian home energy monitoring start-up, Smappe is estimated to have recorded a CAGR of more than 50% since 2015 in the home energy management systems (HEMS) market. The company's products are compatible with those offered by most of the connected home solution providers such as Google, Amazon, Varta, Niko, Netatmo, Daikin, Tado, and Signify. The company's products support IFTTT, Conrad Connect and MQTT, OCPP, Modbus, Bluetooth, KNX and more protocols.	The AI-based home energy monitoring device detects every appliance for energy consumption and generation through the incorporation of machine learning algorithms. The home energy monitoring system consists of hardware devices for tracking electricity, gas, and water consumption. It is installed in the breaker panel or on appliances to track energy consumption and energy anomalies of appliances via a dashboard application.

Source: Frost & Sullivan

Competitive Analysis

Top AI Companies to Watch (continued)

High-performing companies.

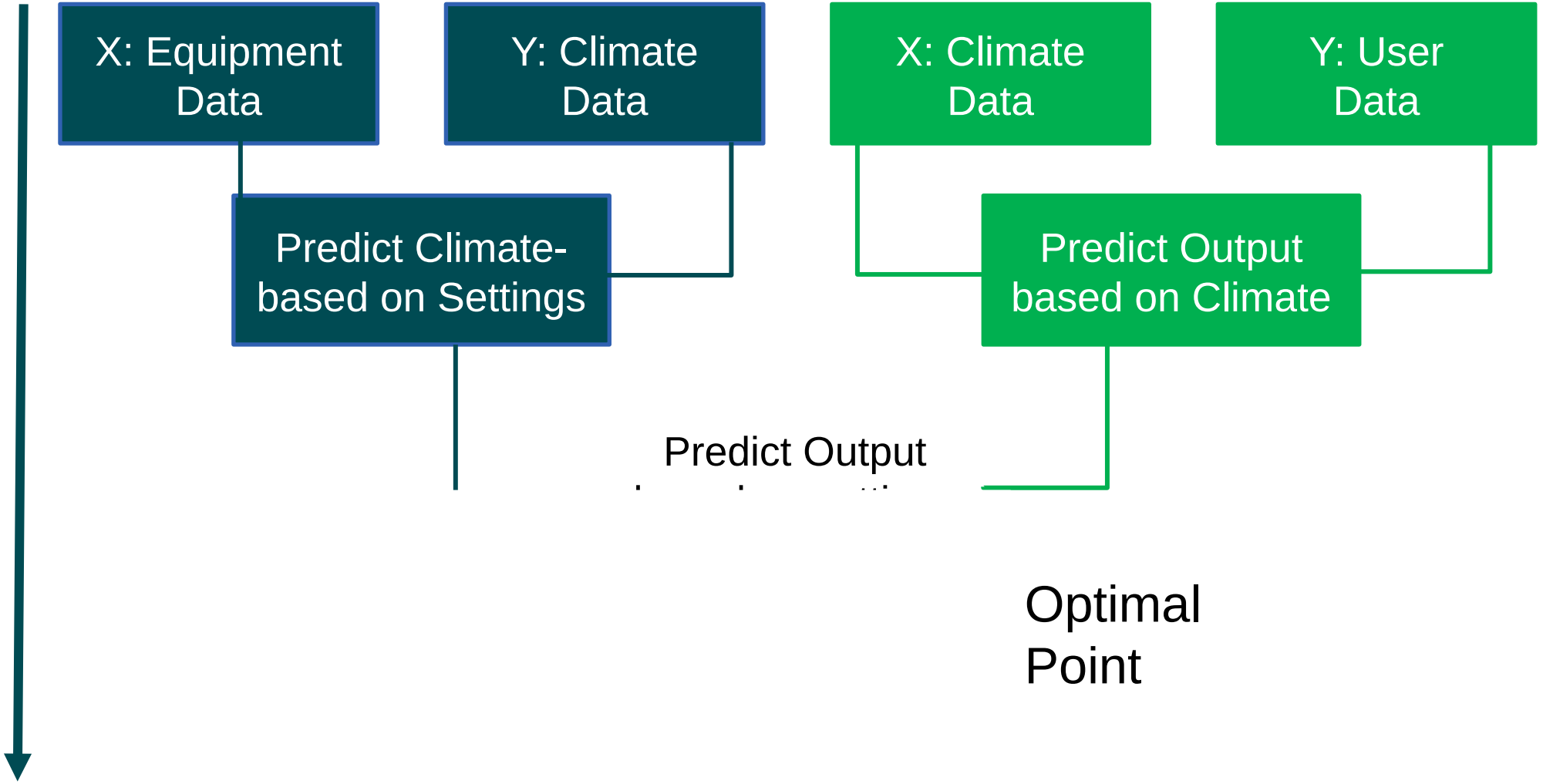
Company	Growth	Innovation
CURB	Energy Curb, a US-based home monitoring start-up, has been growing at an accelerated rate over the past couple of years. The company has partnered with Schneider Electric to integrate its home energy monitoring product with Schneider's Square D smart load breaker panel, a modular product. This partnership has helped Schneider Electric enter the home energy management market and allowed Energy Curb to expand its customer base.	The AI-based home energy monitoring device detects every appliance by incorporating machine learning algorithm. The product is a hardware device installed in the breaker panel and a mobile application to track energy consumption and energy anomalies of appliances.
VELUX	The Danish company is one of the leaders in roof windows, skylights, and intelligent home controls. In 2017, the company partnered with Netatmo to offer an indoor climate control solution to its residential customers. Frost & Sullivan considers it as a highly impactful, value-driven, AI-based solution available in the residential market.	Velux ACTIVE is an indoor climate control solution that includes a Wi-Fi gateway and smart controls for Velux INTEGRA roof windows and sunscreening products. The solution incorporates a machine learning algorithm for ventilation and blind control.
VATTENFALL	Vattenfall continues to invest significantly in renewable production. An investment of about $2.46 billion is planned for wind farms, of which 75% is for offshore wind farms. The company is also investing $100 million in solar and battery projects and another $170 million in distribution network solutions and decentralised heat solutions. The company has bought 100% shares of Senfal and is adding new services for industrial demand-side flexibility and renewable generation optimisation.	Vattenfall's acquisition of Senfal has made it one of the very few companies offering autonomous demand response and power trading solutions. Senfal's AI algorithm enables renewable energy operators and large industrials to optimally use their assets, and the platform makes autonomous decisions on energy markets.

Source: Frost & Sullivan

Underlying magic
(unfair advantage)

Competitive Advantage: Computer System for Management of Building Equipment

Singapore patent pending: 10202105064T



Marketing / Distribution Plan how to acquire customers?

Marketing / Distribution Plan